

Supermarket Edge AI CCTV System

Hardware Sizing, Calculations & Materials Checklist

Low-Cost Architecture: Intel Core i5 | RX 9060 XT (16GB) | 16GB DDR5 | 512GB SSD + NAS Storage | 32 Channels

Deployment Architecture: 32 IP CCTV Cameras (Aisles, Fresh, Bakery, POS, Entrances, Liquor & Logistics)

Core Edge AI PC: Intel Core i5-13400F/14400F | AMD Radeon RX 9060 XT (16GB GDDR6) | 16GB DDR5-5200 | 512GB NVMe SSD

Existing NAS Offload: Continuous 24/7 video loop streamed directly to existing store NAS via NFS/SMB (saves ~\$125 on local SSDs)

Material Classification: **[MANDATORY]** required for base system operation; **[OPTIONAL]** marked depending on whether existing supermarket equipment (router, UPS, cameras, rack) is retained.

1. Executive Summary & Cost Benefits

This specification establishes the mathematical sizing, memory allocations, and verified component requirements for deploying a cost-optimized 32-camera edge AI video analytics system. By shifting continuous 24/7 video loop recording to the supermarket's existing Network Attached Storage (NAS) server and adopting the Intel Core i5 paired with the AMD Radeon RX 9060 XT (16GB GDDR6), the total capital expenditure (CAPEX) for the compute appliance is reduced by over 50% (from ~\$2,000 down to ~\$833–\$953 USD) while retaining the 16GB VRAM mandate, deterministic hardware video decoding, and real-time vision analytics without operational compromise.

2. Engineering Calculations & Workload Sizing (32 Cameras)

- **Network Ingress Bandwidth:** 32 Sub-Streams (720p @ H.265 / 2.0 Mbps = 64.0 Mbps) + 32 Main-Streams (1080p @ H.265 / 6.0 Mbps = 192.0 Mbps) = 256.0 Mbps (~32.0 MB/s continuous). On a 2.5 GbE interface, this utilizes only ~10.2% wire capacity, absorbing synchronized I-frame bursts (500–600 Mbps) with zero packet jitter.
- **Hardware Video Decoding (AMD VCN):** 32 cameras decimated to 3 FPS for AI inference = 96 FPS aggregate. Dedicated AMD Video Core Next (VCN 4.0/5.0) silicon decodes >260 FPS of 1080p or >650 FPS of 720p H.264/H.265, operating at only ~15% to 18% load, leaving massive headroom for incident zooms.
- **GPU VRAM Allocation (16GB GDDR6):** Video decode DPB buffers (1.10 GB) + Tensor staging & scaling (0.90 GB) + Multi-model AI weights [YOLOv8, Pose, ReID] (2.80 GB) + ROCm driver

context & MIOpen cache (2.30 GB) + Local 8B Ollama LLM (5.50 GB) + Safety headroom (3.40 GB) = 16.0 GB total.

- **Host System RAM (16GB Minimum):** Headless Linux OS (1.20 GB) + Edge backend core (0.90 GB) + 32-stream RTSP TCP/UDP socket buffers (1.40 GB) + SQLite WAL & DuckDB cache (1.00 GB) + Temporary frame IPC (0.70 GB) + Linux page cache (2.20 GB) = ~7.40 GB active (~54% free safety headroom on 16GB RAM).
- **Local NVMe Storage (512GB SSD):** Linux OS & Docker (25 GB) + Python toolchain (15 GB) + AI models & LLM weights (20 GB) + SQLite DB (30 GB) + Emergency 35-minute failover ring buffer (50 GB) + Wear-leveling free space (~372 GB, >72% free, guaranteeing 10+ years NVMe lifespan).
- **NAS Storage Throughput & Retention:** Continuous write stream: 24.0 MB/s (~86.4 GB/hr = ~2.07 TB/day). 7-Day retention requires ~14.5 TB; 14-Day requires ~29.0 TB; 30-Day requires ~62.0 TB RAID storage on existing NAS. Utilizes only 21% of a 1 GbE port or <8% of a 2.5 GbE port.

3. Itemized List of Materials Required (BOM)

3.1 Edge AI Compute Appliance (The Edge PC)

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[MANDATORY]	Dedicated GPU	AMD Radeon™ RX 9060 XT (16GB GDDR6, 128-bit) [Alt: RX 7600 XT 16GB]	1	\$330 – \$360
[MANDATORY]	Host Processor (CPU)	Intel Core i5-13400F or i5-14400F (10 Cores, 16 Threads, 65W TDP)	1	\$165 – \$190
[MANDATORY]	Motherboard	ASRock B760M Pro RS or MSI PRO B760M-A (Micro-ATX, 2.5GbE LAN, Dual M.2)	1	\$110 – \$130
[MANDATORY]	System Memory (RAM)	16GB (2x 8GB) DDR5-5200 MHz CL36 Dual-Channel Kit (Kingston/Crucial)	1	\$55 – \$65
[OPTIONAL]	RAM Capacity Upgrade	32GB (2x 16GB) DDR5-5600 CL30 (Enables 10GB RAM-disk pre-alarm buffer)	1	(+\$35 upgrade)

[MANDATORY]	Local Storage (SSD)	512GB WD Blue SN580 or Crucial P3 Plus PCIe 4.0 M.2 NVMe SSD	1	\$40 – \$48
[MANDATORY]	Power Supply (PSU)	650W 80+ Bronze/Gold (MSI MAG A650BN / Corsair CX650)	1	\$55 – \$65
[MANDATORY]	Computer Chassis	Montech Air 100 Mesh or DeepCool CC360 (Micro- ATX, 4x Fans, High- Airflow)	1	\$45 – \$55
[MANDATORY]	CPU Cooler	Thermalright Assassin X 120 Refined SE (120mm Silent Tower Cooler)	1	\$18 – \$20
[MANDATORY]	Secondary Network Card	PCIe 2.5 GbE Network Card (Realtek RTL8125B) for CCTV isolated VLAN	1	\$15 – \$20

3.2 Network Infrastructure (Switching & Routing)

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[MANDATORY]	48-Port PoE+ Managed Switch	TP-Link Omada TL-SG3452P (48x GbE PoE+, 384W power budget, 4x SFP)	1	\$480 – \$520
[OPTIONAL]	Commercial 2.5GbE Router	TP-Link Omada ER707-M2 (2x 2.5GbE, 4x GbE, 802.1Q VLANs, Multi-WAN) [Procure if existing router lacks 2.5GbE/VLANs]	1	\$165 – \$190
[OPTIONAL]	Cellular Failover Modem	4G/5G USB Dongle (Plugs into router for automatic internet failover)	1	\$45 – \$70

3.3 Power Protection & Battery Backup (UPS)

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[OPTIONAL]	1500VA Pure Sine Wave	APC Smart-UPS SMT1500IC	1	\$550 – \$620

	UPS	(1500VA/1000W, LCD, USB) [Procure if site lacks 1500VA UPS with USB auto-shutdown]		
[OPTIONAL]	Alternative UPS (Budget)	CyberPower PR1500ELCD (1500VA/1500W Pure Sine Wave, LCD, USB)	1	\$460 – \$510
[MANDATORY]	UPS USB Signaling Cable	Type-A to Type-B Cable (Connects UPS to Edge PC for Linux auto-shutdown daemon)	1	Included in box

3.4 Cameras (32 Channels — Optional if existing cameras are retained)

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[OPTIONAL]	Aisle Cameras (1–12)	Dahua IPC-HDW2441T-S (3.6mm Lens, 4MP Starlight, 120dB WDR, IP67) [Optional if existing cameras reused]	12	\$900 – \$1,020
[OPTIONAL]	Produce & Bakery Cameras	Dahua IPC-HDW2441T-S (2.8mm Wide Lens, 4MP Starlight, 120dB WDR) [Optional if existing cameras reused]	4	\$300 – \$340
[OPTIONAL]	POS Checkout Cameras (1–6)	Dahua IPC-HDW2441T-S (3.6mm/6mm Lens, 4MP @ 30 FPS, Built-in Mic) [Optional if existing cameras reused]	6	\$450 – \$510
[OPTIONAL]	Entrance & Foyer Cameras	Dahua IPC-HDBW2441E-S (2.8mm Wide, 4MP, 120dB WDR, IK10 Vandal Dome) [Optional if existing cameras reused]	4	\$320 – \$360
[OPTIONAL]	Liquor & Dock Cameras	Dahua IPC-HDW2441T-S (Turret) + IPC-HFW2441S-S (Bullet IP67) [Optional if existing cameras reused]	6	\$450 – \$510

3.5 Server Cabinet, Cabling & Installation Accessories

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[OPTIONAL]	Equipment Server Cabinet	9U or 12U Wall-Mount Enclosure (600mm Depth, Glass Door) [Procure if missing]	1	\$140 – \$180
[MANDATORY]	Cat6 48-Port Patch Panel	1U 19" Cat6 110 Punch-Down / Keystone Panel [Procure if missing]	1	\$35 – \$50
[OPTIONAL]	Bulk Cat6 UTP Cable	305m Box Solid Bare Copper UTP 23AWG (2 boxes = 610m) [Procure if rewiring]	2 boxes	\$220 – \$280
[MANDATORY]	Pass-Through RJ45 Plugs	Cat6 Pass-Through RJ45 8P8C Plugs + Relief Boots (100-pack)	1 pk	\$18 – \$22
[MANDATORY]	Rackmount 8-Outlet PDU	1U 19" Horizontal PDU Power Strip with Surge Suppression	1	\$35 – \$45
[MANDATORY]	Cat6 Molded Patch Leads	0.5m & 1.0m Snagless Slim Cat6 Patch Cords (35-pack for switch-to-panel)	1 pk	\$28 – \$35
[OPTIONAL]	Horizontal Cable Manager	1U 19" Finger Duct Cable Panel with Removable Cover	1	\$15 – \$20

3.6 Existing NAS Server Requirements (Zero New Purchase)

Status	Component / Material	Verified Part Specification	Qty	Est. Price (USD)
[EXISTING]	Network Storage (NAS)	Existing Supermarket NAS (Synology, QNAP, TrueNAS, unRAID) via NFS v4 / SMB 3.0	1 unit	\$0.00 (Existing)
[EXISTING]	NAS Storage Pool	RAID-5 / RAID-6 pool with surveillance HDDs (requires ~14.5TB for 7-day, ~29TB for 14-day)	1 pool	\$0.00 (Existing)
[EXISTING]	NAS Network Interface	1 GbE or 2.5 GbE	1 port	\$0.00 (Existing)

Ethernet connection to
core switch (requires ~24
MB/s throughput)

4. Procurement Investment Summary

Category	Scope Included	Estimated Cost (USD)
1. Edge AI Appliance PC [MANDATORY]	Core i5, RX 9060 XT 16GB, 16GB DDR5, 512GB SSD, 650W PSU, Case, Dual NIC	\$833 – \$953 USD
2. Network Infrastructure [MANDATORY/OPT]	48-Port PoE+ Switch (\$480-\$520) [MANDATORY] + Router (\$165-\$190) [OPTIONAL]	\$480 – \$710 USD
3. Battery Backup UPS [OPTIONAL]	1500VA / 1000W Pure Sine Wave Smart UPS with USB auto-shutdown cable	\$460 – \$620 USD
4. Racking & Cabling [MANDATORY/OPT]	Patch panel, PDU, Patch leads, RJ45 [MANDATORY]; Rack & Bulk Cable [OPTIONAL]	\$116 – \$632 USD
5. 32x Surveillance Cameras [OPTIONAL]	32x Dahua 4MP WizSense IP Cameras (Optional if existing cameras are retained)	\$2,420 – \$2,740 USD

TOTAL MANDATORY HARDWARE (Edge AI PC Only): \$833 – \$953 USD

TOTAL INFRASTRUCTURE UPGRADE (PC + PoE Switch + Router + UPS + Rack/Cables): \$2,429 – \$2,915 USD

ALL-INCLUSIVE COMPLETE TOTAL (WITH 32 NEW DAHUA CAMERAS): \$4,849 – \$5,655 USD